

FEATURES

- USB-IF Certified. TID: 11363
- Dual-Role Port PD Compatible
- Attach/Detach Detection as Host, Device or DRP
- Current Capability Definition and Detection
- Cable Recognition
- Dead Battery Support
- Ultra-low Power Mode for Attach Detection
- Simple I²C Interface with Indication Pin
- Dual Slave Address for Dual Port Application
- BIST Mode Supported
- Programmable Default Settings
- Supported PD 3.0 except Fast Role Swap Function and Unchunked Message
- 9-Ball WLCSP (WLCSP-9B) and 14-Lead QFN (QFN-14L) Packages

APPLICATIONS

Smartphones

Tablets
Laptops
Monitors

GENERAL DESCRIPTION

The **HUSB311** is a USB Type-C PD controller that complies with the latest USB Type-C and PD standards. It implements the USB Type-C CC control and sensing and USB PD Message delivery. The **HUSB311** has programmable R_p and R_d settings for each CC line. It does the USB type-C detection including attach and orientation. The **HUSB311** integrates a complete BMC encoding including a receiver and transmitter. With this physical layer of the USB BMC power delivery protocol, the **HUSB311** is able to handle the PD protocol and support any power up to 100W and perform role swap as needed. The BMC PD block enables full support for alternative interfaces of the Type-C specification.

The **HUSB311** uses I²C to communicate with the TCPM via employing an INT signal for requesting attention.

TYPICAL APPLICATION CIRCUIT

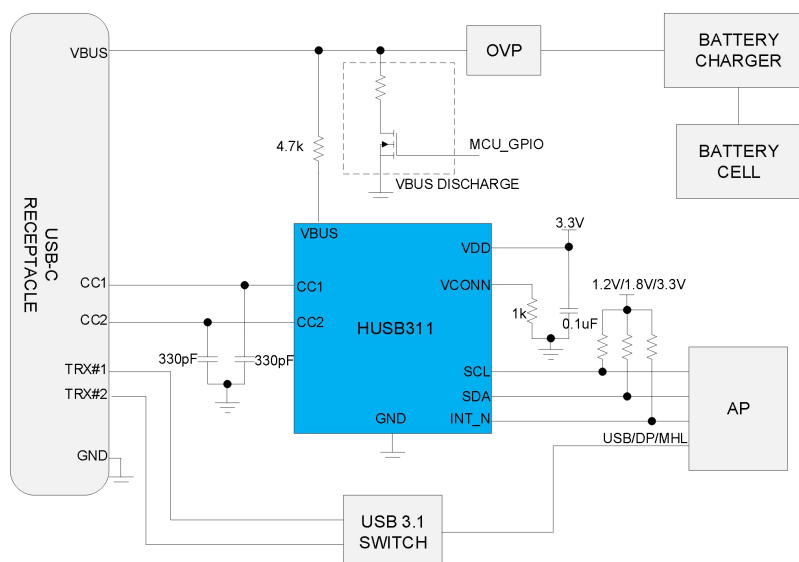


Figure 1. HUSB311 Typical Application Circuit